One-plus is absorbed exactly above omega

The Open Logic Project contributors

openlogic:set-theory/ord-arithmetic:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ord-arithmetic/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the five labels ord:notinomega/ord:omegaplus/ord:oneplus/ord:plusone/ord:infinite share one statement and one proof; this card counts the shared proof under ord:oneplus (the constructive hinge) and the sibling labels are not separately counted (same span, one card per LANE_BRIEF OLP same-span rule)

Editorial note. The label ord:oneplus names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Five-way infinity characterisation for ordinals. The lemma ordinalinfinitycharacter states the equivalence of five conditions: alpha not in omega, omega <= alpha, 1+alpha = alpha, alpha ~ alpha+1, and alpha Dedekind infinite.. The counted card is the omega<=alpha -> 1+alpha=alpha implication, the constructive hinge of the equivalence cycle; the cycle's other arrows are one-step citations or short bijection constructions delivered in the same proof.

Source reference. Original theorem naturalnumbersarentinfinite: sources/olp-set-theory-z/pinned-original-source.tex, source lines 534–536. Cross-unit citation invoked in the final implication of the counted proof: the target label is staged in the corpus under unit olp-set-theory-z at source lines 534-536; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *For any ordinal α, the following are equivalent:*

1. *$\alpha \notin \omega$, i.e., α is not a natural number*
2. *$\omega \leq \alpha$*

3. $1 + \alpha = \alpha$

4. $\alpha \approx \alpha + 1$*, i.e.,* α *and* $\alpha + 1$ *are equinumerous*

5. α *is Dedekind infinite*

Proof. *1* $\Rightarrow$ *2.* By Trichotomy.

2 $\Rightarrow$ *3.* Fix $\alpha \geq \omega$. By Transfinite Induction, there is some least ordinal γ (possibly 0) such that there is a limit ordinal β with $\alpha = \beta + \gamma$. Now:

$$1 + \alpha = 1 + (\beta + \gamma) = (1 + \beta) + \gamma = \subsetneq_{\delta < \beta} (1 + \delta) + \gamma = \beta + \gamma = \alpha.$$

3 $\Rightarrow$ *4.* There is clearly bijection $f \colon (\alpha \oplus 1) \to (1 \oplus \alpha)$. If $1 + \alpha = \alpha$, there is an isomorphism $g \colon (1 \oplus \alpha) \to \alpha$. Now consider $\circ fg$.

4 $\Rightarrow$ *5.* If $\alpha \approx \alpha + 1$, there is bijection $f \colon (\alpha \oplus 1) \to \alpha$. Define $g(\gamma) = f(\gamma, 0)$ for each $\gamma < \alpha$; this injection witnesses that α is Dedekind infinite, since $f(0, 1) \in \alpha \setminus \operatorname{ran} g$.

5 $\Rightarrow$ *1.* This is naturalnumbersarentinfinite. $\square$